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Assessment of wetland and vegetation dynamics in the context of urbanization and seasonal variation using surface indices in Ara city, India

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Abstract

The pressure of rapid urbanisation has led to the rapid transformation of wetlands and vegetation, causing major problems for ecological sustainability in fast-growing cities. This study assesses the spatiotemporal dynamics of wetland, vegetation, and built-up surfaces in Ara City, Bihar, India, from 1995 to 2025 using Landsat 5, 8, and 9 imageries. The following spectral indices were used to assess the seasonal and long-term environmental changes as a result of urban expansion: Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), Modified Normalized Difference Water Index (MNDWI), Normalized Difference Built-up Index (NDBI), Urban Index (UI), Normalized Difference Bareness Index (NDBal), Bareness Index (BI), and Enhanced Built-up and Bareness Index (EBBI). The anthropogenic impact on vegetation and water resources around Ara city in middle of Bihar-Uttar Pradesh border in Bhojpur district of north-eastern part of Bihar, India surrounding by the Ganga and Son rivers was studied. The results reveal a progressive decline in vegetation cover and open-water features, areas experiencing rapid urban expansion. Pre-monsoon NDVI exhibited lower spatial variability than post-monsoon NDVI, indicating seasonal differences in vegetation dynamics. MNDWI indicated a reduction in water-related features, whereas higher NDWI values in 2025 suggest an increase in water-associated surfaces within the urban landscape. The results of the Pearson's correlation analysis indicate negative correlation between built-up indices and NDVI and negative correlation between built-up indices and MNDWI, showing the inverse relationship between urbanization and ecological health indicators, while positive correlation between built-up indices and NDWI shows the positive relationship between built-up and water. The study underscores that wetlands, which serve as critical habitats for migratory birds, are increasingly threatened by human activities. The novel nature of the study is the integration of multi-season and multi-index analyses of the dynamics of wetland, vegetation, and urban surfaces over 30 years. The results offer insights for sustainable urban planning, wetland conservation, ecosystem management and may have implications for wetland-dependent biodiversity.



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Keywords Urbanization, Surface indices, Wetland, Vegetation, Migratory birds

1 Introduction

Urban encroachment reduces vegetation and wetland coverage in urban environments. The wetland and the vegetation coverage are two significant components of our ecosystem. The first one acts as a kidney [5] and another one acts as a water purifying system and green lung [30]. Both of these are habitats for many living species on our planet. Man-made activities that gain onto the natural environment form a fragmented network of different organs in our ecosystem. For the betterment of our future generation in the context of contemporary resource utilization, a blueprint was prepared, known as the Sustainable Development Goal [41]. Urban greening is negatively redistributive in global cities [15].

Urban land alteration is a major threat to global terrestrial biodiversity; losses of native biodiversity abundance and species abundance are estimated at 34% and 52% per 1 km grid-cell respectively, and an average loss of 7–9 species per 10 km grid-cell [22]. In this context, open spaces in urban built-up areas contribute a critical part to regulating the microclimate, primarily in the regulation of urban temperature [33]. Deforestation and urban gentrification caused environmental contamination to spread quickly throughout Asia, which may have long-term effects on sustainable development and economic growth [3]. Land use/land cover (LULC) modifications have been accelerated by rapid urbanization, as evidenced by a 9.36% uplift of built-up zones that affects land surface temperature (LST), a 5.73% reduction of cropland, and diminishing vegetation cover by 3.56%, as a result of decreased Normalized Difference Vegetation Index (NDVI) values and a push-up to the urban heat island (UHI), as well as a reduction of the anthropogenic factor of the COVID-19 lockdown in 2020, which temporarily increased NDVI values and decreased LST [17]. Numerous habitats, such as parks, cemeteries, urban forests, vacant lots, campus areas, gardens and yards, and landfills, as well as blue spaces like lakes, streams, artificial swales, ponds, and storm water retention ponds, generate urban ecosystem services. It is predicted that between 2000 and 2030, over 60% of the urban land that will exist in 2030 will be constructed [11, 31]. The urban “green and blue infrastructure” has a critical contribution towards enhancing adjustment capability and survival strength in contemporary climatic uncertainty situations and economically long-lasting, multiservice civilian life in our society [12, 19]. The key contribution of flora and fauna in urban areas is innumerable. The open green and blue space has a center of aesthetic demand and recreational needs of maximum urban citizens. In a deeper sense, these elements serve important functions such as replenishing groundwater, reducing pollution, setting pollution benchmarks, controlling climate effects and flood damage, providing shade for vendors and travelers, and mitigating heat during the sunny season [10].

According to MoEFCC, wetlands have the following values and benefits, e.g., wetlands as sources of water, wetlands as flood and storm buffers, wetland products, wetlands as water purifiers, wetlands for recreation and tourism, wetlands and climate change through CO₂ sinks, and wetlands as habitats for migratory birds and biodiversity hotspots [27]. Rural-to-urban migration leads to expanded anthropogenic activities,

which pressure local biodiversity, ecosystem services, and natural hazards and require ethical teaching, training, and awareness programs to preserve these urban organs [6].

Although remote sensing techniques have been used to monitor urban environmental changes and significant progress has been made, the existing studies have focused mainly on one type of urban features at a time (e.g., vegetation, wetlands, built-up areas), and/or included only a few spectral indices or a limited number of temporal observations. The comprehensive assessment of vegetation, wetland, barren and built-up land dynamics over several seasons and over a longer time scale has thus been constrained, especially for medium sized cities in eastern India that are rapidly urbanizing. In addition, few studies have quantitatively compared the relationships among several spectral indices to increase an understanding of the ecological response of urban landscape to rapid urbanization. Ara city, situated between Ganga and Son rivers has marked urbanization in the past few decades, but there is a lack of complete geospatial assessment of the wetland and vegetation dynamics of the city. The origins of the name Ara were believed to come from “Aaranno,” which means forest or vegetated area, which is reflecting the historically close connection of the region to the presence of plenty of vegetation and wetlands. These ecosystems have traditionally been able to sustain the local diversity of wildlife and can offer appropriate habitat for wetland dependent species, such as migratory birds. The present study aims to mitigate the identified research gap by using multi-temporal Landsat data (1995–2025) and eight spectral indices (NDVI, NDWI, MNDWI, NDBI, UI, NDBal, BI and EBBI) to explore the long-term and seasonal evolution of wetlands, vegetation, barren and built-up land. Pearson’s correlation analysis is also utilized to further measure relations among these indices and assess their responses to urbanization. The integrated multi-season, multi-index approach used in this study is novel and offers a holistic view of long-term urban ecological changes, which will generate scientific evidence to guide sustainable urban planning, wetland conservation and ecosystem management. Changes in wetlands and vegetation observed are also discussed in terms of their potential ecological impact on wetland dependent biodiversity, including migratory bird habitat, although no direct impacts are assumed in the absence of field-based ecological observations.

1.1 Objective

1. To evaluate the spatiotemporal variation of wetland and vegetation cover from 1995 to 2025.
2. To examine the impact of urbanization on wetland and vegetation cover and discuss its potential implications for migratory bird habitats.
3. To identify the relation of different surface indices from 1995 to 2025.

Hypothesis (H0) There is no correlation of different surface indices from 1995 to 2025.

2 Datasets and procedures

2.1 Selection of the research region

Ara city existing headquarter of the district Bhojpur in Bihar, is the only class-1 town of the district and is situated at the height of 192 m from sea level with the confluence of Sone and Ganges River. Geographically, Ara lies between latitudes 25° 10' and 25° 40' N and longitudes 83° 45' and 84° 45' E, with a distance of 58 km from the Bihar state

capital of Patna (Fig. 1). The relative position of the city of Ara, situated 13 km west of the Sone River and 22 km south of the Ganges, is vulnerable to flooding during the monsoon season due to the northern region being low-lying and close to the Ganges River. Ara is densely populated, forming a triangular area between the Ganges and the Ara Canal, bordered to the north and northwest by Sobhi Dumra, Cheetarsenpur, and Sandiya; to the east by Piprahiyan and Bhushaula; and to the south by Dariyapur, Chakiya, and Ekauna. Surrounding districts include south Rohtas, east Patna, Jehanabad, Arwal, north Saran (Bihar), Balia (U.P.), and west Buxar. Ara is the administrative headquarters of Bhojpur district in Bihar. According to the 2011 Census, it is among the major urban centres of the state and covers an area of approximately 30.97 km² under the Ara Municipal Corporation. Ara Nagar Panchayat was established in the year 1929 and in 2002, it was renamed as Nagar Parishad and later as Nagar Nigam in 2007. After the 2007 ward boundary reorganisation, there are 45 wards in the municipality compared with 32 wards recorded in the 2001 Census. Ara is the administrative city of Bhojpur, with several offices working in the city such as water resources, public health, engineering, transport, and agriculture. It is also home to sub-courts and the District Sessional Court, which serve the judiciary in the region, highlighting its significance as a judicial and administrative hub.

2.2 Growth of population

From the year 1901 to 1911, Ara city has experienced a significant decline of population, attributed fundamentally to the impact of the “Plague,” which guided to a 16.5% decrease during this period. In the year of 1921 has been identified as the Demographic Divide,

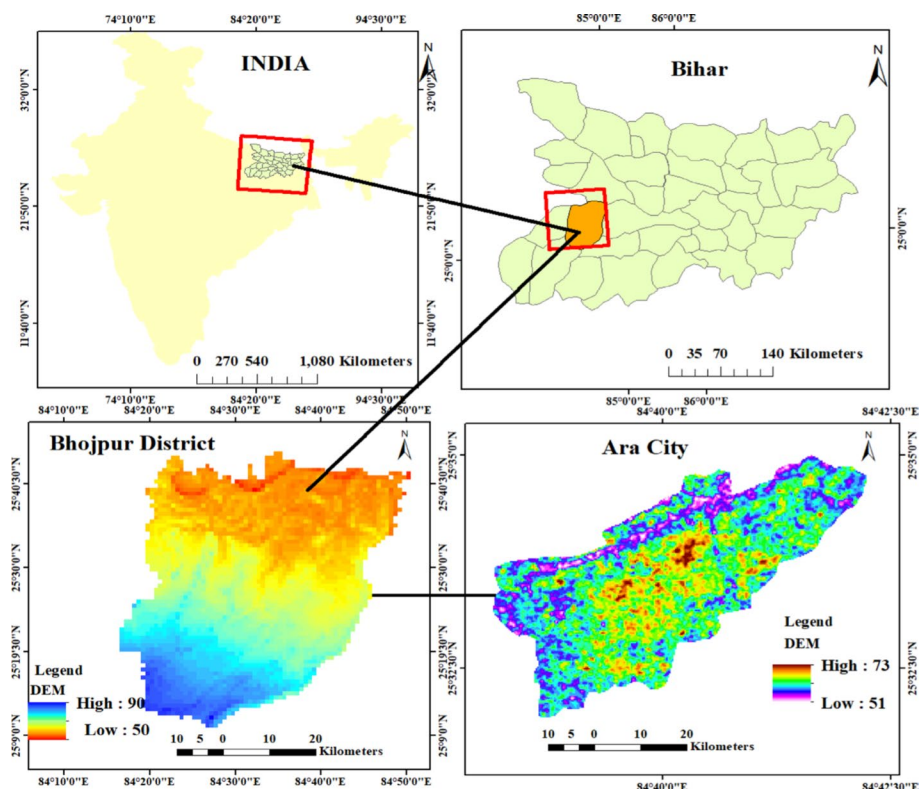


Fig. 1 Map of research interest region

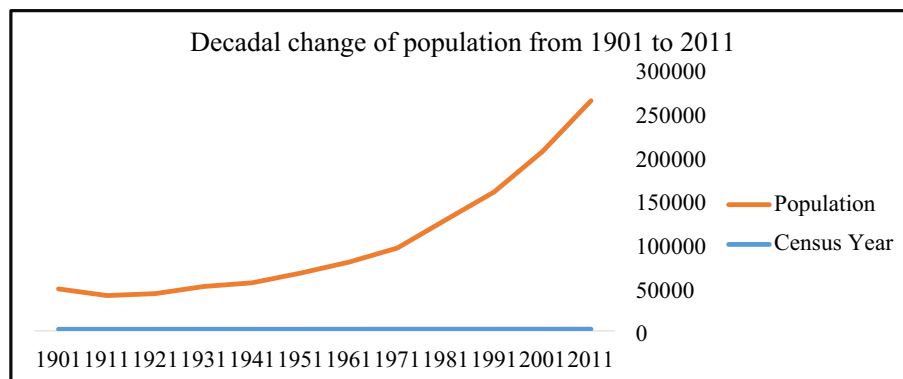


Fig. 2 Trend of decadal population growth

Table 1 Decadal variation of population of Ara city. Source: District Gazetteer and Census Book of the mentioned years (1901–2011)

Census year	Population	Variation in decadal growth
1901	46,170	–
1911	38,549	– 7621
1921	40,769	+ 2220
1931	48,922	+ 8153
1941	53,122	+ 4200
1951	64,205	+ 11,083
1961	76,766	+ 12,561
1971	92,919	+ 16,153
1981	125,111	+ 32,192
1991	157,082	+ 31,760
2001	203,380	+ 46,298
2011	261,430	+ 58,050

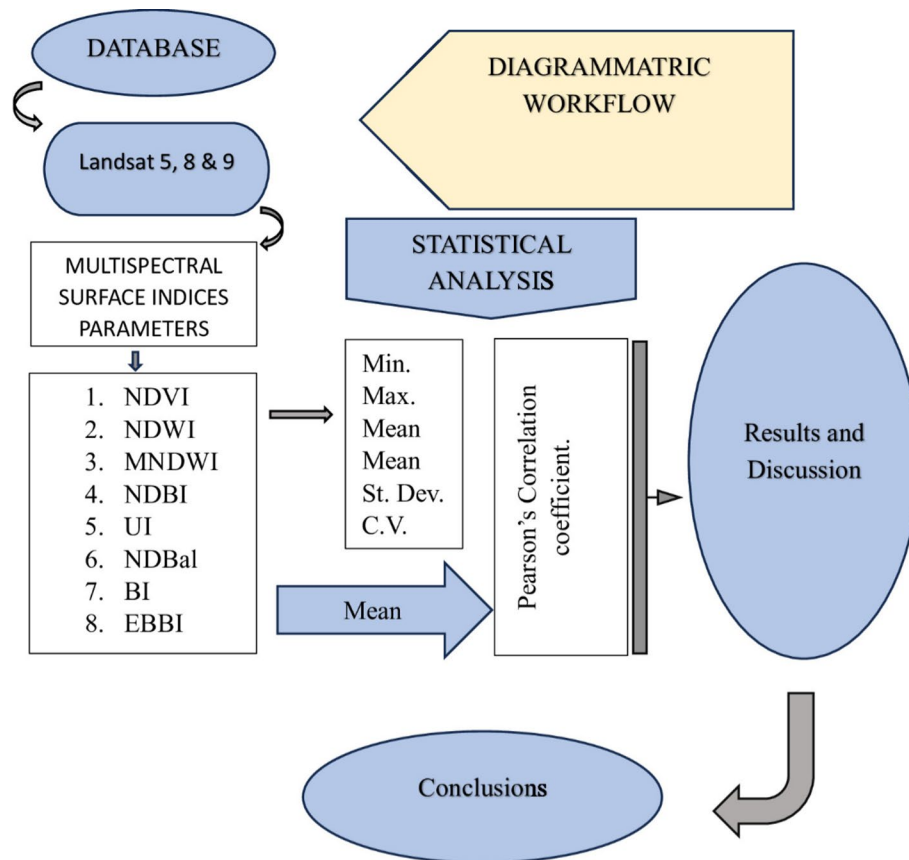
marking the only census year within this timeframe from 1901 to 2011 that recorded a slowdown in population growth. Impressively, while the overall population in India was decreasing, Ara was unique as it disclosed both an increase and a decrease in its population. Specifically, the population rose from 38,549 in 1911 to 40,769 in 1921, achieving a decadal growth rate of 5.8% (Fig. 2). Following this period, the population of Ara city has continued to grow interruptedly. The data regarding these changes is further illustrated in Table 1, which presents Ara city's decadal growth rates.

2.3 Database

Landsat TM second generation is widely used than MSS for its higher spatial and spectral resolutions (30 m). Landsat satellites have ideal spectral bands and ground resolution for monitoring purpose of land occupation and recording dynamics of land cover as a consequence of urbanization, deforestation, climatic fluctuation, and multi- human and natural-caused phenomena[17, 39, 46]. The US Geological Survey has supplied an optical multispectral and thematic mapper for download (<https://earthexplorer.usgs.gov>) [42]. Table 2 shows details of different satellite images, such as acquisition date, sensor types, processing correction level, and spatial resolution. Figure 3 shows diagrammatic workflow of the methodological framework adopted for the study.

Table 2 Specification of satellite images. Source: Landsat Handbook

Sensor and satellite	Date of acquisition	Processing correction level	Resolution in space (m)	Row/path
Landsat TM 5 (LT05)	23/04/1995	L2C2	30 & 60	042/141
Landsat TM 5 (LT05)	01/11/1995	L2C2	30 & 60	042/141
Landsat OLI (LC 09)	05/05/2025	L2C2	30 & 60	042/141
Landsat OLI (LC 08)	18/10/2025	L2C2	30 & 60	042/141

**Fig. 3** Diagrammatic workflow of methodology

2.3.1 Satellite data selection, image preprocessing, and data processing workflow

The Multi-temporal Landsat satellite imagery was retrieved from the United States Geological Survey (USGS) Earth Explorer archive to monitor the changes in wetland, vegetation and built-up surfaces in Ara city between 1995 and 2025. The Landsat 5 TM imagery was used for 1995 while the Landsat 8 OLI/TIRS and Landsat 9 OLI-2/TIRS-2 imagery was used for 2025. Images were obtained in April-May (pre-monsoon) and October-November (post-monsoon) to reflect the seasonal variations in the study area. The pre-monsoon period is comparatively a dry period with lesser vegetation greenness and surface water availability, while the post-monsoon period follows the southwest monsoon, during which vegetation is at its peak growth and wetlands are refilling. The comparison of these two seasons will allow for a full evaluation of seasonal changes in vegetation, surface water and urban land cover changes and will reduce the impact of short-term climatic variability. Those images that lack a maximum cloud cover (below 10%) were chosen to minimize atmospheric interference. Geometrically and

radiometrically corrected Level-2 surface reflectance products were downloaded from USGS. Further pre-processing consisted of clipping the imagery to the boundary of the administrative area of Ara city, checking spatial alignment and masking non-study area pixels in ArcGIS 10.8 software. Standard mathematical expressions were used in ArcGIS 10.8 to generate surface spectral indices (such as NDVI, NDWI, MNDWI, NDBI, UI, NDBal, BI, and EBBI). The obtained raster layers were classified and analyzed to assess changes in vegetation, wetlands, built up and bare ground on a spatial and temporal basis. The average pixel value of each spectral index was computed and Pearson's correlation analysis was conducted at the 95% confidence level ($p \leq 0.05$) using Microsoft Excel 2021. The overall workflow consisted of (i) satellite image acquisition, (ii) image preprocessing, (iii) generation of spectral indices, (iv) spatial analysis and mapping, (v) extraction of statistical parameters, and (vi) correlation analysis and interpretation of long-term environmental changes.

2.3.2 Vegetation index

Normalized difference vegetation index (NDVI)

For vegetation analysis, the NDVI is a globally recognized spectral index [7, 20]. Because growing plants retain more red light and reflect near-infrared radiation, first developed the NDVI equation, or the ratio of red to near-infrared radiation in 1974 [34]. Water bodies, bare terrain, and populated places are indicated by values between zero and negative, and positive values show thin to thick vegetation cover. NDVI levels vary from -1 to 1 [25].

$$\text{NDVI} = ((\text{NIRr} - \text{REDr})) / ((\text{NIRr} + \text{REDr})) \quad (1)$$

REDr and NIRr indicate the red and near-infrared band reflectance values, whereas NDVI stands for normalized difference vegetation index.

2.3.3 Water indices

Normalized difference water index (NDWI)

The NDWI is a useful spectral index for delineating open water body aspects developed by McFeeters [26]. Shortwave-infrared (SWIR) and near-infrared (NIR) radiation are absorbed by water bodies. NIRr and Greenr are the reflectances of the near-infrared and green bands, respectively, and McFeeters defined it as

$$\text{NDWI} = (\text{GREENr} - \text{NIRr}) / (\text{GREENr} + \text{NIRr}).$$

First NDWI was introduced in 1996, a remote sensing technique to estimate the canopy level vegetation water content from space using SWIR and NIR bands [13]. SWIR and Red bands (bands 5 and 3 in Landsat TM) were employed by Rogers and Kearney to generate NDWI [18]. The surface is covered with non-water bodies if $\text{NDWI} \leq 0$, while it is filled with water if $\text{NDWI} > 0$. The NDWI value ranges from -1 to 1 , with zero serving as a threshold value. In this research, the Eq. 2 has been used for NDWI calculation [26]. i.e.,

$$\text{NDWI} = ((\text{GREENr} - \text{NIRr})) / ((\text{GREENr} + \text{NIRr})) \quad (2)$$

where NIRr and GREENr represent the reflectance values of the near-infrared and green bands, respectively, and NDWI stands for normalized difference water index.

Modified normalized difference water index (MNDWI)

The combination of open water bodies and ponds is shown by the MNDWI values. A normalized difference pond index (NDPI) has been invented in 2007 to categorize ponds in West Africa [21]. Compared to the NDWI, the MNDWI shows more open water and is more suited for improving water with a lot of built-up land in the background [44]. Water bodies and built-up objects cannot be completely distinguished by McFeeters's NDWI. This problem was addressed in 2006, and the Eq. 3 has been used for calculation of MNDWI [44]. The modified normalized difference water index is known as MNDWI. This equation was further used in 2007 [21, 23].

$$\text{MNDWI} = ((\text{GREENr} - \text{SWIRr})) / ((\text{GREENr} + \text{SWIRr})) \quad (3)$$

where GREENr and SWIRr represent the reflectance values of the green and short-wave infrared bands, and

2.3.4 Built-up and bareness indices

There are many indices for analysis of built-up and bareness areas. This study has calculated NDBI, EBBI, UI, and BI for comparative analysis of different built-up and bareness indices on urban density measures. Each of them has an owned equation, i.e.,

Normalized difference built-up index (NDBI)

For mapping built-up regions automatically, NDBI was first developed by a Chinese geographer, in 2003 [45]. Thereafter, it was frequently used to calculate the built-up area of cities. Water bodies are indicated by a negative NDBI value, built-up areas are indicated by a greater positive NDBI value, and vegetation coverage is indicated by a lower positive NDBI value. NDBI is an outcome of the transformation of NDVI for analyzing built-up region; thereafter, it has been used for the intensity of urban construction [28]. The equation follows for calculating NDBI.

$$\text{NDBI} = ((\text{SWIR1r} - \text{NIRr})) / ((\text{SWIR1r} + \text{NIRr})) \quad (4)$$

where SWIR1r and NIRr represent the reflectance values of the short-wave infrared/middle-infrared, and near-infrared channels of various bands in Landsat imagery, and NDBI stands for normalized difference built-up index.

Urban index (UI)

$$\text{UI} = ((\text{SWIR2r} - \text{NIRr})) / ((\text{SWIR2r} + \text{NIRr})) \quad (5)$$

where UI means urban index: SWIR2_r, and NIR_r represent mid-wave infrared channel 2 and near infrared band.

Normalized Difference Bareness Index (NDBal)

$$\text{NDBal} = ((\text{SWIR1r} - \text{TIRr})) / ((\text{SWIR1r} + \text{TIRr})) \quad (6)$$

where NDBal means normalized difference bareness index: SWIR1_r, and TIR_r represent short wave infrared channel 1 and thermal infrared band.

Bareness Index (BI)

$$\text{BI} = [((\text{SWIR1r} - \text{NIRr})) / ((\text{SWIR1r} + \text{NIRr}))] - [((\text{NIRr} - \text{REDr})) / ((\text{NIRr} + \text{REDr}))] - \text{or } \text{NDBI} - \text{NDVI} \quad (7)$$

where SWIR1r, NIRr, and REDr stand for short wave infrared band channel 1 reflectance, near infrared, and red band, respectively; BI stands for bareness index.

Enhanced built-up bareness index (EBBI)

$$EBBI = ((MIRr - NIRr)) / (10\sqrt{(MIRr + TIRS1r)}) \quad (8)$$

where EBBI means enhanced built-up bareness index; MIR_r, NIR_r, and TIRS1r subsequently represent mid-infrared, near infrared and thermal infrared (B10) band.

Pearson's correlation coefficient

After processing and extracting the satellite image, different statistical values such as maximum, minimum, mean, coefficient of variation, and standard deviation have been calculated, and at the end of the study, Pearson's correlation coefficient will be calculated in all parameters during the study period from 1995 to 2025.

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}} \quad (9)$$

3 Results

To examine vegetation, wetland, and built-up region with the help of different surface indices, conducted with respect to surface indices fulfilling NDVI, NDWI, MNDWI, NDBI, UI, NDBal, BI, and EBBI with the help of multispectral images from different years. Figure 4a–h (i–iv) visualized the year-wise maps for different surface indices. Details of the processed image have been represented in the given table (Tables 3 and 4). Quotative measures are more reliable than qualitative (visualize data) interpretations of data [25].

3.1 Vegetation coverage analysis

The mean value of vegetation indices showed the pre-monsoon: 0.130196 in 1995 and 0.164198 in 2025, based on the study area's total pixel count of 20,203 and cell size (x, y): 29.89507876, 29.89507876. The standard deviation, which indicates the concentration of data around the mean value, is in 1995 shown as 0.041664 and in 2025 shown as 0.053105. A fragile relation shows with the mean when the standard deviation is higher. A pre-monsoonal study reveals that the standard deviation in 1995 is lower than in 2025, which indicates that the vegetation coverage has changed and fragmented from 1995 to 2025 in the pre-monsoon season. After the monsoon period, the mean of vegetation indices is 0.29579 in 1995 and 0.212437 in 2025. The standard deviation in 1995 is 0.050176, and in 2025, it is 0.081453. Statistical analysis indicates that, based on standard deviation, the vegetation index value in 1995 is more concentrated around the mean than in 2025 during the offset of monsoon period. The post-monsoon season has shown a similar pattern to pre-monsoon.

Deep green to yellow-red hues represented the spatiotemporal variation of forest depth from highest to lowest with regard to NDVI (Fig. 4a (i–iv)). The NDVI's minimum (Min.) and maximum (Max.) values during the post-monsoon season in 1995 were greater than the pre-monsoon Min. and Max. values (Table 3). These Min. and Max. values indicate the success rate of vegetation health is higher in post-monsoon than pre-monsoon arrival season due to an abundance of precipitation, humidity, sunshine, and

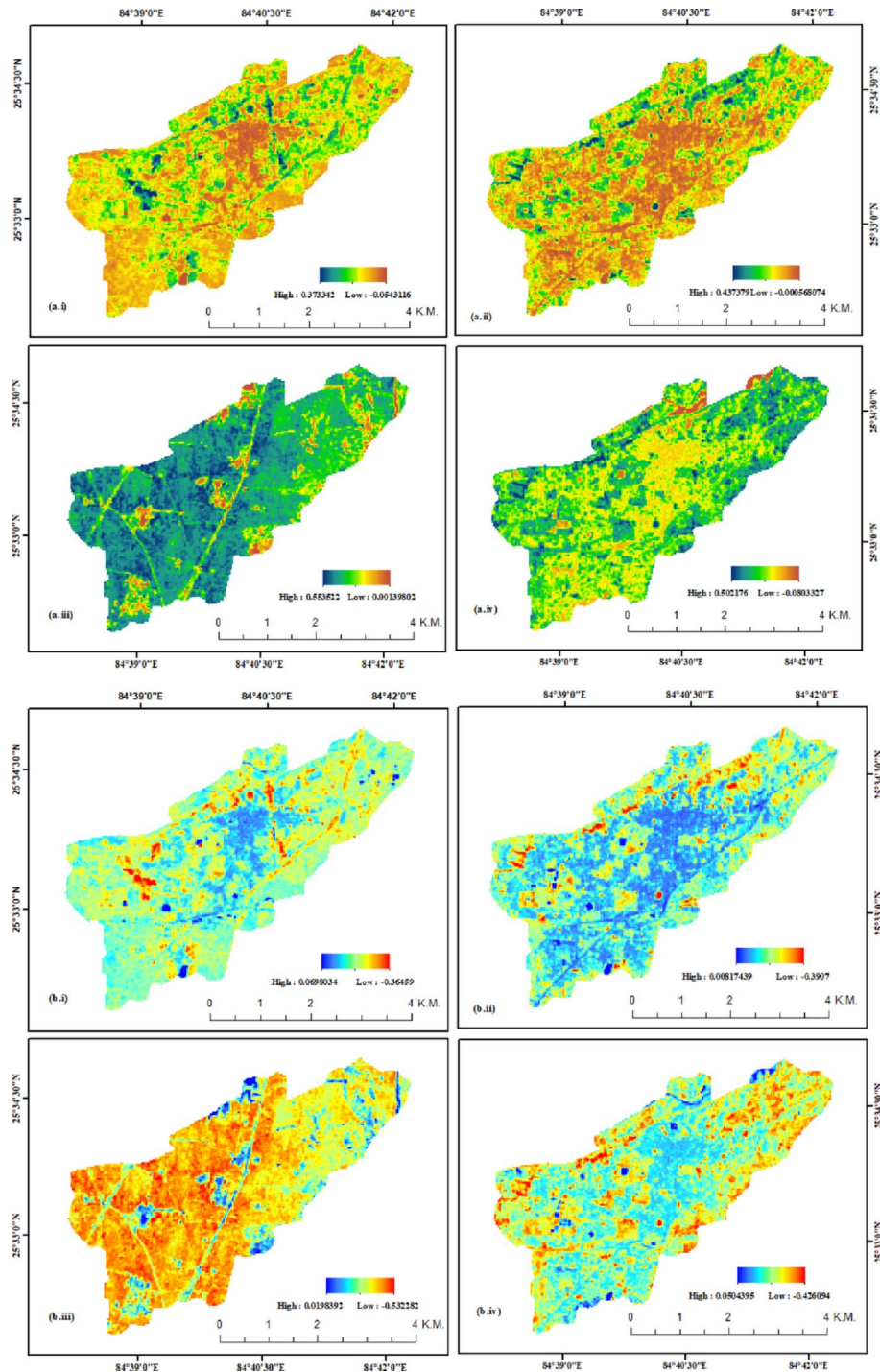


Fig. 4 Status of various surface indices through spectral indices value. **a** Normalized Difference Vegetation Index (NDVI), **b** Normalized Difference Water Index (NDWI), **c** Modified Normalized Difference Water Index (MNDWI), **d** Normalized Difference Built-up Index (NDBI), **e** Urban Index (UI), **f** Normalized Difference Bareness Index (NDBal), **g** Bareness Index (BI) and **h** Enhanced Built-up and Bareness Index (EBBI). N.B. in figures, (i) stands for April 1995(pre-monsoon season), (ii) stands for May 2025 (pre-monsoon season) and figures, (iii) stands for November 1995(post-monsoon season), (iv) stands for October 2025(post-monsoon season)

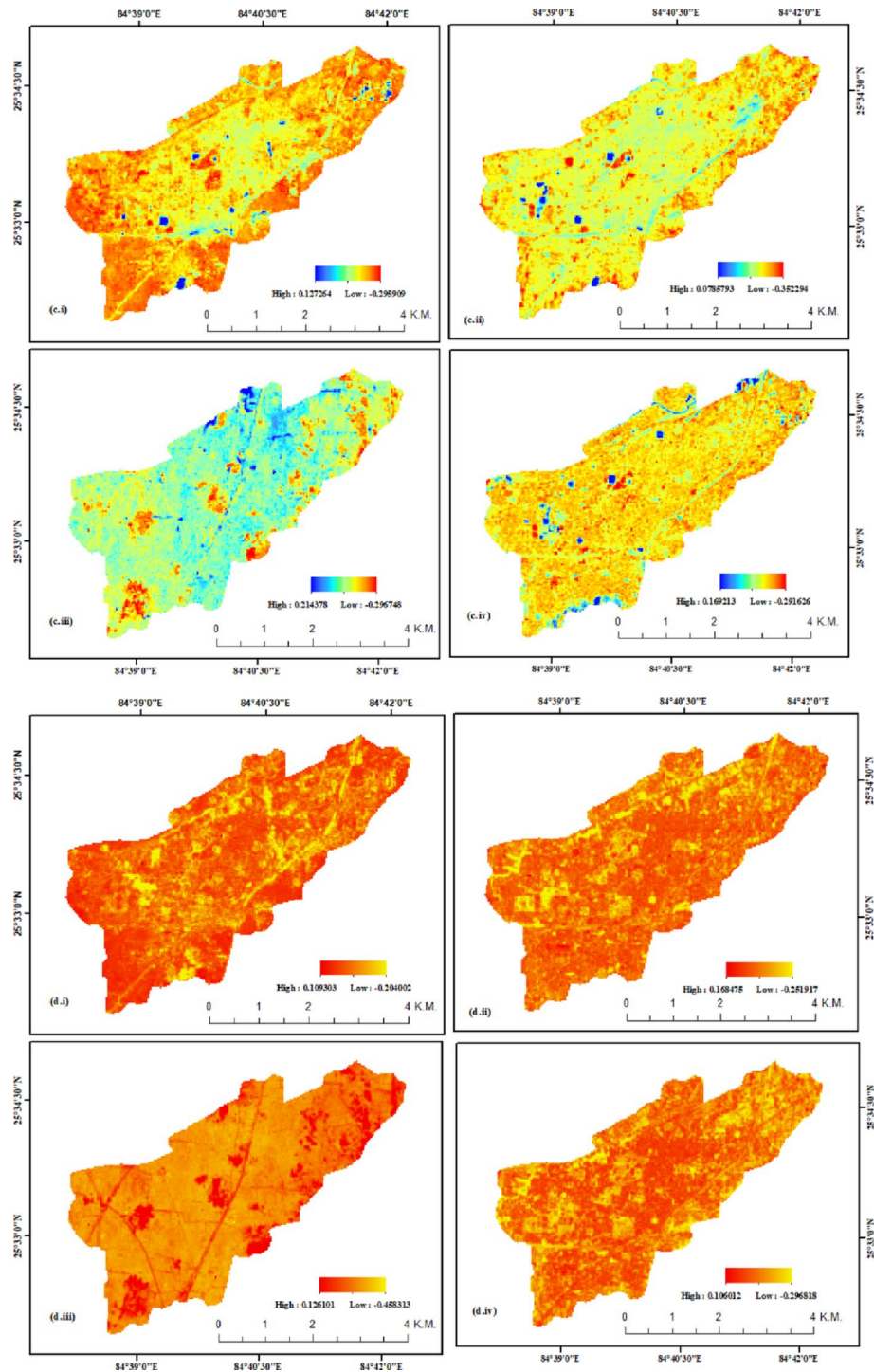


Fig. 4 (continued)

other relative factors that help vegetation growth. Regarding the years 1995 to 2025, the minimum value of -0.426096 (post-monsoon) and the maximum value of 0.050440 (post-monsoon) suggest that the wider range observed in the post-monsoon season indicates a modification in a factor that has negatively affected vegetation health over this period. The vegetation health status is better in 2025 than it was in 1995, with a minimum value of -0.000568 and a maximum value of 0.437379 in pre-monsoon scenarios.

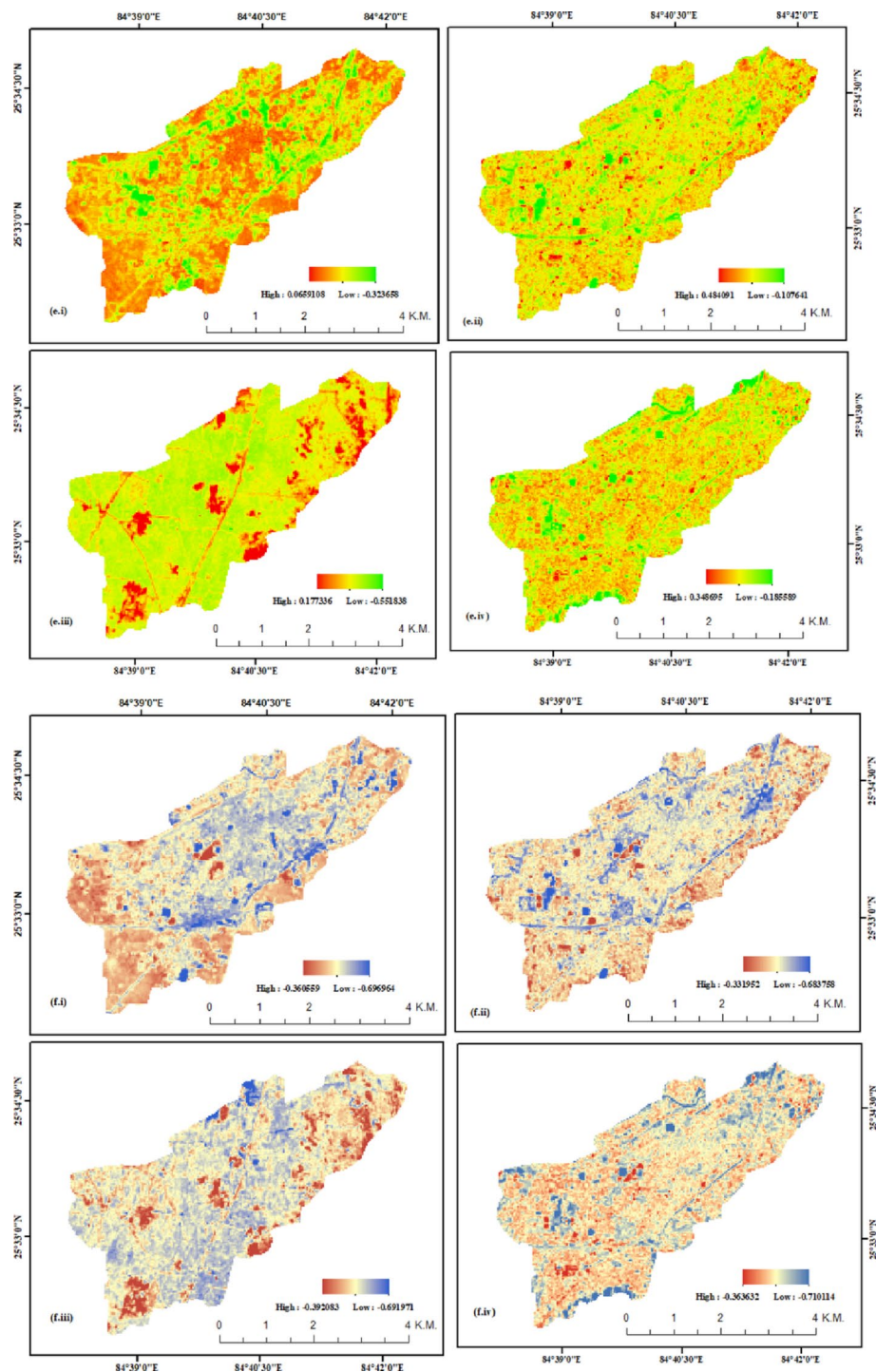


Fig. 4 (continued)

The negative value, which is below the threshold of 0, is also lower than in 1995 during pre-monsoon conditions. This suggests that human efforts to nurture nature are making progress, which is commendable for environmental conservation.

Ara city's vegetation health is clearly good, as evidenced by the improvement in NDVI value and consistency of pixel values during the pre-monsoon season over time. However, in the post-monsoon case, there is an inverse correlation, which suggests that both

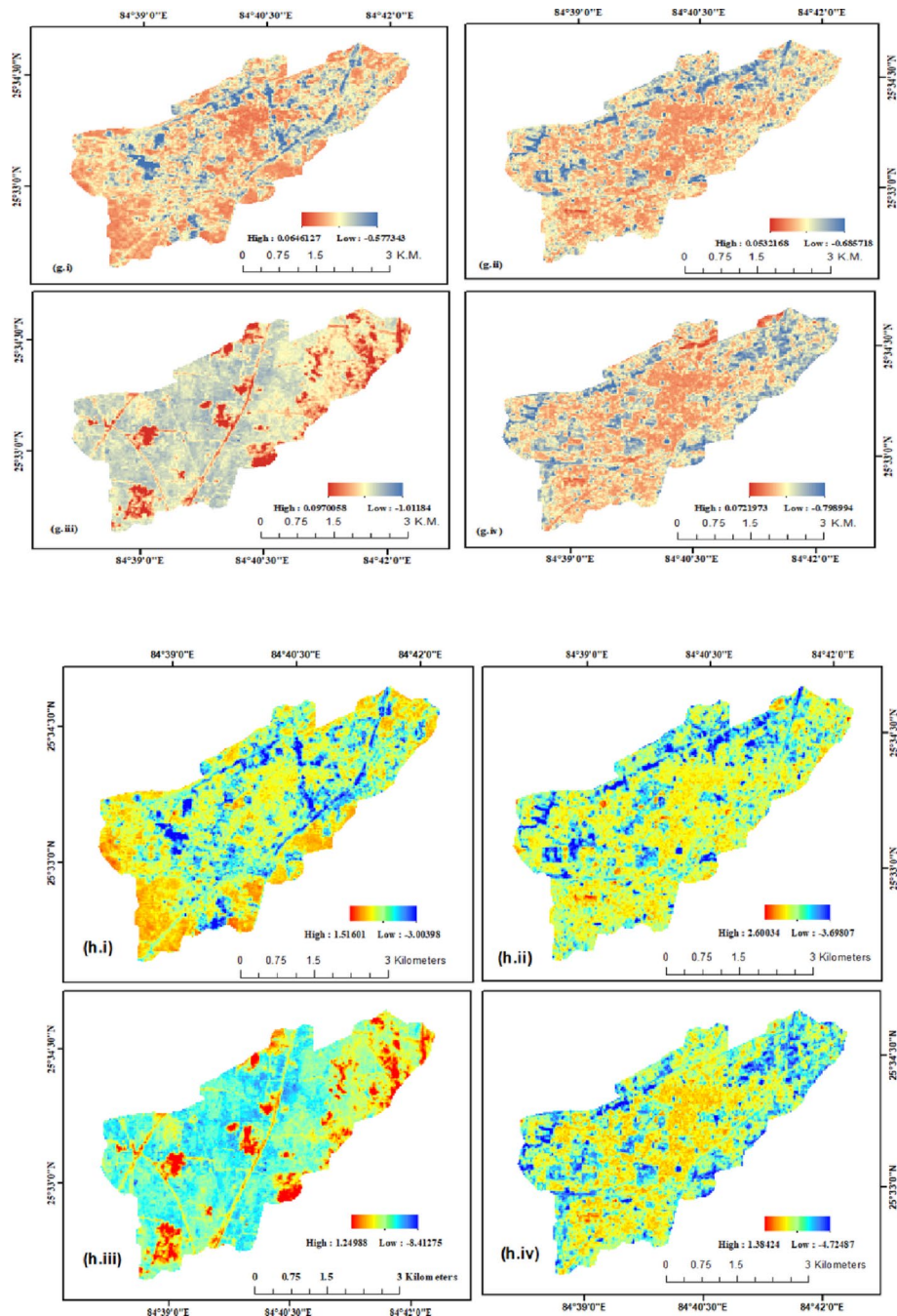


Fig. 4 (continued)

human and climatic factors broke up the dense vegetation patch and overstressed its growth. This is the reason why, after the monsoon, the vegetation cover does not achieve the index value in the preceding year, i.e., in 1995. Although consistency is somewhat better (32.0010%) in 1995 than (32.3421%) in 2025, the coefficient of variation (C.V.) displayed a variable pattern; mean values increased in pre-monsoon season in both 1995 and 2025. After overall analysis of NDVI, high consistency has been found in November 1995, indicating that healthy vegetation existed more predominantly earlier than in the contemporary situation, and the 'homeostasis' capacity of vegetation was higher than

Table 3 Statistical analysis of different surface indices. Source: Computed by researchers

Surface spectral indices		Year	Pre-monsoon (Apr.–May)					Post-monsoon (Oct.–Nov.)				
			Min.	Max.	Mean	St. Dev.	C.V. (%)	Min.	Max.	Mean	St. Dev.	C.V. (%)
Normalized Difference Vegetation Index (NDVI)		1995	−0.05431	0.373342	0.130196	0.041664	32.001	0.001398	0.553522	0.29579	0.050176	16.96338
		2025	−0.00057	0.437379	0.164198	0.053105	32.3421	−0.4261	0.050044	0.212437	0.081453	38.34219
Normalized Difference Water Index (NDWI)		1995	−0.36459	0.0698	−0.16246	0.03512	−21.6171	−0.53228	0.01984	−0.28159	0.04118	−14.6221
		2025	−0.3907	0.00817	−0.18222	0.04371	−23.9849	−0.42609	0.050044	−0.20144	0.05841	−28.9977
Modified Normalized Difference Water Index (MNDWI)		1995	−0.2959	0.127264	−0.18446	0.035351	−19.1638	−0.29674	0.214378	−0.12099	0.031303	−25.8706
		2025	0.35229	0.07858	−0.17905	0.03005	−16.7811	−0.29162	0.16921	−0.15081	0.03723	−24.6854

Table 4 Statistical analysis of different surface indices. Source: Computed by researchers

Spectral surface indices		Pre-monsoon (Apr.–May)		Post-monsoon (Oct.–Nov.)							
		Min.	Max.	Mean	St. Dev.	C.V. (%)	Min.	Max.	Mean	St. Dev.	C.V. (%)
Normalized Difference Built-up Index (NDBI)	1995	−0.204	0.10931	0.02267	0.04152	183.154	−0.45831	0.1261	−0.1662	0.05685	−34.2062
	2025	−0.25192	0.16848	−0.00347	0.04651	−1342.14	−0.204	0.10601	−0.05255	0.06336	−120.588
Urban Index (UI)	1995	−0.32366	0.06591	−0.04192	0.05464	−130.35	−0.55184	0.17734	−0.28164	0.06887	−24.4544
	2025	−0.10764	0.48409	0.09464	0.0269	28.4193	−0.18559	0.3487	0.08647	0.03371	38.9833
Normalized Difference Bareness Index (NDBaI)	1995	−0.69696	−0.36056	−0.49558	0.036732	−7.4119	−0.69197	−0.39208	−0.55714	0.02697	−4.84079
	2025	−0.68376	−0.33195	−0.4927	0.02944	−5.9747	−0.71011	−0.36363	−0.5335	0.03518	−6.59493
Bareness Index (BI)	1995	−0.57734	0.06461	−0.10753	0.07764	−72.203	−1.01184	0.09701	−0.46199	0.10401	−22.5144
	2025	−0.68572	0.05322	−0.16766	0.09567	−57.0585	−0.79899	0.0722	−0.26498	0.13643	−51.4873
Enhanced Built-up and Bareness Index (EBBI)	1995	−3.00398	1.51601	0.29661	0.52805	178.03	−8.41275	1.24988	−2.10832	0.75001	−35.5736
	2025	−3.69807	2.60034	−0.04485	0.61026	−1360.73	−4.72487	1.38424	−0.64736	0.79857	−123.358

in the current situation. In the research area, the compared index map (Fig. 4a (i–iv)) in the pre-monsoon season showed compacted green patches in 1995 in west-northwest (WNW) region of study, but in 2025, compacted green patches showed in north-north-east (NNE) region of study, whereas deep reddish-yellow patches showed in the middle place of the region of study in 1995, but in 2025, this has spread from the middle to western parts of the region of study.

Post-monsoon season of 1995, the region of study area was mostly green, which suggested good and dense vegetation. A smaller portion in pivotal, western and northern regions, however, had a compact reddish-yellow colour, indicating relatively low vegetation health. This changed considerably by 2025, when the reddish-yellow colour covered a much larger area than in 1995 at the time of the measurements. This shift suggests a deterioration in the health of vegetation over time, especially the pre-monsoon and post-monsoon periods, from 1995 to 2025. These images can clearly be visually interpreted to reveal the spatio-temporal dynamics of the study area's changing land usage.

3.2 Analysis of water bodies

In the study region, there are not many water bodies. The water and modified index of water are used for calculating open-water features and water bodies surrounding built-up areas. Based on data extracted from satellite imagery, the pre-monsoonal mean of the water index value is less negatively distributed than the post-monsoonal (Table 3). In pre-monsoon, the mean of the water index is -0.162464 in 1995 and -0.182223 in 2025. The standard deviation of the pre-monsoon water index in 1995 is 0.035120 , and in 2025 it is 0.043706 . After the monsoon, it has been replaced; in 1995, it was 0.041175 , and in 2025, it is 0.058413 . For NDWI, more consistency was found in 1995 (pre-monsoon and post-monsoon) than in 2025 (pre-monsoon and post-monsoon). In pre-monsoon period between 1995 and 2025, healthier water bodies were present earlier than in the contemporary situation. In post-monsoon season, it showed a drastic change between 1995 and 2025; a higher value was found in 2025 than in 1995. This season, the water index pixel values are more consistent in 1995 than in 2025. The standard deviation is lower in 1995 than in 2025, and the C.V. shows the data are more consistent in 1995 than in 2025. This indicates that the spatial variability of deepwater bodies has increased due to human interventions, such as landfills meeting the higher demand for valuable urban land compared to the urban periphery for their building construction, but consistency of data reveals that there are insufficient restorations of wetlands compared to earlier times.

The index map of NDWI, deep blue colour showed water bodies, and light blue colour showed wet agriculture fields. Reddish-orange and yellow showed vegetation, natural fallows, built-up areas, etc. In the study region, the index map (Fig. 4b (i–ii)) for the pre-monsoon season showed that the light blue colour was more scattered in 1995 than in 2025. In 2025, it showed wider, smoother expansion of blue colour, but after analysing the 20,203-pixel value of the satellite image, the arithmetic data of the NDWI mean was -0.162464 in 1995 and -0.182223 in 2025, which indicates that the 1995 mean value is closer to the threshold value (i.e., 0) than the 2025 mean value. The mean values, standard deviations, and CVs represented in 1995 were of greater consistency than in 2025, and the water body distribution was smoother than in 2025. The index map (Fig. 4b (iii–iv)) after the monsoon season showed that deep blue colour in the north, middle, and

west middle in 1995; a more natural shape indicates that there is less anthropogenic factor that has modified the natural environment than in 2025. In 2025, deep blue showed round and square shapes that indicate the human intervention has changed the water bodies on a temporal basis in the region of study.

The standard deviation of the modified water index is lower in 2025 than in 1995, and C.V. showed more consistency in 2025 than in 1995. This indicates the increase in human activities, and as a consequence, man-made water bodies have been accelerated. Urbanization has been utilizing soil as a raw material for increasing the height of roads, dwellings, and lower swamps and wetlands in the heart of the city for enhanced dwelling height. As a consequence, open-water features are reduced due to landfill; open water bodies increase due to the extraction of soil for maintaining dwelling height from surrounding areas. On the other hand, the modified water index standard deviation in 1995 is 0.031303, and in 2025 it is 0.037229 in the post-monsoon season; consequently, C.V.-expressed data are more consistent in 2025 than in 1995. A higher value of pre-monsoon showed 0.127264 in 1995 and 0.078579 in 2025, as well as post-monsoon; it has 0.21438 in 1995 and 0.16921 in 2025. It indicates wetland and water bodies are wider expanses surrounding urban areas in 2025 than in 1995, but health is the severest from 1995 to 2025. These statistics express that, due to urban encroachment, low-lying land areas of the city are utilized for urban building, and the modified water index slightly changes in pre- and post-monsoon compared to the water index because water-associated surfaces have more seasonal change than enclosed wetland and water bodies lying behind or in the center of the city. These water bodies continuously recharge through sewage water. In the context of the modified water index, the pre-monsoonal mean showed -0.184468 in 1995 and -0.179047 in 2025; more consistency was shown in 2025 (i.e., -16.7811) than in 1995 (i.e., -19.1638). Sewage water, climate change, and government policy to conserve wetlands have been leading factors in showing greater consistency in MNDWI in 2025 than in 1995. In the post-monsoon cases, the mean showed -0.120998 in 1995 and -0.150814 in 2025; more consistency was shown in 2025 (i.e., -24.685374) than in 1995 (i.e., -25.870676). After overall analysis, a high level of consistency was observed in the pre-monsoon season in May 2025; the C.V. value is lower (-16.7811 in May 2025) than other C.V.s, i.e., (-19.1638 in April 1995), (-25.870676 in November 1995), and (-24.685374 in October 2025), indicating that due to landfills, water associated surfaces in urban areas transfer to open water masses, water of sewage helps to maintain their areal status, but the health of the water bodies significantly reduces in the pre-monsoon from 0.126274 in 1995 to 0.0785793 in 2025 and in the post-monsoon from 0.214378 in 1995 to 0.169213 in 2025. In this regard, a more remarkable decision will be required for their conservation in Ara city.

The index map of MNDWI, deep blue colour, showed water bodies, and the rest of the colour showed things other than water bodies, like green cover, construction areas, bare land, etc. (Fig. 4c (i–iv)). In 1995, fewer water bodies were visualized in more natural shapes than in 2025, pre-monsoon (Fig. 4c (i–ii)). In 2025, the shapes of water holding surface features changed, and the sharpness of the boundaries of water holding features increased. The post-monsoon season in 1995 water bodies shows a more natural shape and wider extension than in 2025 (Fig. 4c (iii–iv)). In 2025, in the north-northeast (NNE) and south-southwest (SSW) portions of the map, several man-made water bodies are seen after the monsoon. After rigorous study of these images and the arithmetic data

of NDWI and MNDWI, it is indicated that the sharpness and area of water bodies have been modified throughout time. Man is an active agent for this kind of change.

3.3 Analysis of built-up and bareness surface areas

Five spectral indices were calculated to evaluate the built-up and bareness characteristics of Ara city, namely NDBI, UI, NDBal, BI, and EBBI (Table 4; Fig. 4d–h (i–iv)). Overall, the indices show significant urbanization and land use change from 1995 to 2025 both in pre-monsoon and post-monsoon period. During this 30-year period, the Urban Index (UI) recorded the highest growth, from -0.0419 in 1995 to 0.0946 in 2025 during the pre-monsoon season and from -0.2816 to 0.0865 in the post-monsoon season (Table 4), suggesting significant expansion of built-up areas in the 30 years. The statistical measures also show changes in the spatial distribution of built up and barren surfaces. UI has lower standard deviation in 2025 than in 1995 in both seasons, indicating greater consistency of urban growth across space. Likewise, the variation of NDBal was relatively low in the pre-monsoon season of 2025, suggesting that the distribution of barren land related to construction work and land excavation was more uniform. During the post-monsoon season, greater variability is observed, with the seasonal influence of rainfall causing temporary flooding and changing the spectral properties of the low-lying barren surfaces (Table 4). These temporal changes are also reflected in the spatial distribution of the built-up and bareness indices (Fig. 4d–h (i–iv)). The built-up area grew continuously from the urban center to the outer areas from 1995 to 2025 with an increasing concentration of barren land around construction sites and new developments at the outskirts of the city. The spatial patterns suggest that the land surface has been significantly altered by rapid urbanization, characterized by the conversion of vegetation and wetlands with impervious surfaces and bare soils. Overall, the built-up and bareness indices show continuous increase in urbanization and land-cover changes in Ara city during the last 30 years (Table 4; Fig. 4d–h (i–iv)). The combined use of multiple indices provides complementary evidence of increasing urbanization, expansion of barren land, and the associated alteration of the urban landscape.

4 Correlation of different indices means from 1995 to 2025

Pearson's correlation analysis was used in Microsoft Excel 2021 to analyze the correlation of NDVI with NDWI, MNDWI, NDBI, UI, NDBal, BI, and EBBI. The analysis was done using the mean value of each spectral index, obtained from the 20,203 pixels of each of the four satellite images (April 1995, November 1995, May 2025, and October 2025). Hence, the number of observations (n) in the statistical sample used for the correlation analysis was four (4). Pearson's correlation coefficients (r) and related p -values were calculated and p -values < 0.05 were considered statistically significant at the 95% confidence level.

NDVI showed a negative correlation with NDWI between 1995 and 2025 ($r = -0.98310$ and p value $= 0.17101$), but this relation is not statistically significant. In the case of MNDWI, it is positively correlated with NDVI from 1995 to 2025 ($r = 0.990377$ and p value $= 0.00431$). Except for MNDWI, all other indices were negatively correlated with the NDVI index mean value, and among them, only NDWI and NDBal were statistically insignificant, indicating that the p -value exceeded the 0.05 cutoff; the rest of them were statistically significant (Table 5). For NDWI, correlation with MNDWI, NDBI, UI,

Table 5 Correlation coefficient by Pearson; mean value of individual surface indices from 1995 to 2025. Source: Computed by researchers

	NDVI	NDWI	MNDWI	NDBI	UI	NDBal	BI	EBBI
NDVI								
p-value	–							
Pearson's r	–							
NDWI								
p-value	0.17101	–						
Pearson's r	–0.98310	–						
MNDWI								
p-value	0.00431	0.01211	–					
Pearson's r	0.99038	–0.96490	–					
NDBI								
p-value	0.00140	0.00044	0.00044	–				
Pearson's r	–0.99383	0.99546	–0.98551	–				
UI								
p-value	0.02790	0.00979	0.00639	0.38681	–			
Pearson's r	–0.69444	0.79862	–0.70190	0.76979	–			
NDBal								
p-value	0.06309	0.14193	0.00747	0.05582	0.34250	–		
Pearson's r	–0.95584	0.91439	–0.98693	0.94812	0.66485	–		
BI								
p-value	0.00426	0.00520	0.00072	0.00426	0.37783	0.00067	–	
Pearson's r	–0.99821	0.99127	–0.98928	0.99868	0.73609	0.95315	–	
EBBI								
p-value	0.00149	0.00036	0.00048	0.34764	0.64602	0.00022	0.00162	–
Pearson's r	–0.99334	0.99623	–0.98403	0.99996	0.77202	0.94537	0.99844	–

Values are Pearson's correlation coefficients (r), with p -values shown in parentheses. $n=4$, representing the four seasonal observations (April 1995, November 1995, May 2025, and October 2025). Correlations were calculated using the mean index values extracted from 20,203 pixels for each image in Microsoft Excel 2021. Statistical significance was evaluated at $\alpha=0.05$. $p<0.05$ indicates statistical significance

NDBal, BI, and EBBI: only MNDWI is negatively correlated; the rest of them are positively correlated, and only NDBal is not statistically significant. For MNDWI, correlation with NDBI, UI, NDBal, BI, and EBBI—all of them are negatively related and statistically significant. NDBI and the rest of the indices have positively correlated, but NDBal and BI are statistically significant, whereas UI and EBBI failed to prove the relationship. In the urban index with NDBal, BI, and EBBI, a positive relation was shown, but all of these are statistically insignificant to prove a relation. For NDBal, the UI and EBBI index have had a highly positive correlation and a statistically significant region. UI and EBBI showed a positive relation and lay at a significant level. Table 5 shows the details of Pearson's correlation coefficient values of different indices' mean values in Ara city.

5 Seasonal analysis of different indices

The results showed the C.V. of different surface indices of the seasons pre-monsoon and post-monsoon from 1995 to 2025. The NDVI value is greater and more consistent in 1995 pre-monsoon (32.0020%) and post-monsoon (16.96%) than in 2025 pre-monsoon (32.3421%) and post-monsoon (38.342190%) seasons. In between the pre-monsoon seasons, 1995 had greater consistency than 2025, and in between the post-monsoon seasons, 1995 had greater consistency than 2025. The Normalized Difference Water Index (NDWI) showed in 1995 higher consistency pre-monsoon (–21.6171) and post-monsoon (–14.6221) than in 2025 pre-monsoon (–23.9849) and post-monsoon (–28.9977),

but for the MNDWI in 2025, there was greater consistency pre-monsoon (-16.7811) and post-monsoon (-24.6853) than in 1995 pre-monsoon (-19.1638) and post-monsoon (-25.8706). The NDBI showed higher consistency in 1995 pre-monsoon (183.1539) and post-monsoon (-34.2062) than in 2025 pre-monsoon (-1342.1356) and post-monsoon (-120.5880), and the urban index (UI) in pre-monsoon showed greater consistency in 2025 (i.e., 28.4293) than in 1995 (i.e., -130.3497) and, inversely, in post-monsoon, i.e., in 1995 (-24.4544) and 2025 (38.9832). The Normalized Difference Bareness Index (NDBal) showed greater consistency in 2025 (-5.975) than in 1995 (-7.412) in the pre-monsoon season; post-monsoon retreated, 1995 showed higher consistency, -4.8407 , than in 2025, i.e., -6.5949 . The bareness index (BI) in pre-monsoon season exhibited greater consistency in 2025, with a value of -57.059 , compared to 1995, which had a value of -72.203 . However, in the post-monsoon season, the trend reverses; in 1995, the value was -22.5143 , whereas in 2025, it was -51.4872 . The EBBI was more stable in both seasons for 1995 (pre-monsoon and post-monsoon) as compared to 2025 (pre-monsoon and post-monsoon) seasons (178.030 and -35.573612 respectively, in 1995; -1360.734 and -123.3583 respectively, in 2025). The analysis suggests that seasonal variations in several control parameters—such as temperature, precipitation, and agricultural activities—have influenced the surface indices. Additionally, the NDVI is positively correlated with the MNDWI, which indicates the richness of aquatic plants in wetlands surrounding urban areas that are known to regulate aquatic health, while the other indices are negatively related. All indices, except for NDWI and NDBal, have shown statistically significant. In NDBI, UI, and EBBI for built-up analysis, UI index values are more consistent in 2025 than in 1995, which indicates the built-up area changes spatio-temporally. Built-up indices are positively correlated with each other. All built-up indices negatively correlated with NDVI and MNDWI at a $\geq 95\%$ significant level. Only NDWI correlated with all built-up indices positively, with a $p\text{-value} \leq 0.05$.

6 Discussion

The present study shows that there are considerable seasonal and temporal deviations in water, vegetation and built-up surface characteristics in Ara city during the period of 1995 to 2025. The observed deterioration of the vegetation condition in the post-monsoon period, increased fragmentation of green spaces and the expansion of built-up surfaces indicate a substantial alteration of the landscape over the study period. The results are similar to those of previous research, which has shown that urbanisation and land-use/land-cover (LULC) change led to a decline in vegetation cover and changes in urban ecological conditions [17, 29, 35]. The same patterns have been found in recent urban remote sensing studies using long-term satellite observations of urban growth, which suggest that the rapid expansion of an urban area has a significant impact on vegetation distribution and urban landscape fragmentation, but that the extent of the change depends on the local environmental conditions and the type of urban planning [16]. The present results are similar to the studies conducted in Indore [1] and Guwahati [17] where reduction in NDVI due to growing urbanization led to alteration in urban environmental conditions. Similarly, recent research has shown that decreases in urban vegetation often occur simultaneously with increased impervious surfaces and landscape fragmentation in rapidly expanding cities. The findings, however, contrast with the findings reported around the Bandel Thermal Power Station where the reduction in

atmospheric particulate matter after the improvement in environmental conditions was the basis for stating that the vegetation health has improved [25]. The difference indicates that in Ara, a number of factors other than urbanization such as seasonal climatic variability, land-use change, environmental management, and urban development may be affecting vegetation dynamics [16].

Analysis of NDWI and MNDWI further showed temporal variations in surface water features. The relatively greater temporal stability of the MNDWI compared to the NDWI, and the strong negative correlation of MNDWI with built-up indices, are consistent with previous GIS-based analyses of wetland loss due to land filling and urban sprawl [24, 38]. Studies on urban landscape dynamics have also shown that urban growth has an impact on wetland ecosystems, and on the diversity of the landscape [2, 29]. The satellite-based investigations have also revealed how water bodies in urban areas and wetlands are often altered by urban growth, especially in areas of land conversion and infrastructure development, over the past few years [9].

However, the existing study is based on spectral indices derived only from remotely sensed data and these relationships must be viewed with caution. Possible modification of wetlands with urban development is noted in the observed spatial patterns, but hydrological variation, seasonal rainfall variation, alteration of drainage patterns, and local water-management practices may also influence changes observed. The study found that there were strong negative correlations between NDVI and built-up indices (NDBI, UI, BI, and EBBI), which align with other studies that found an inverse relationship between vegetation cover and urbanization [17, 36]. These results have also been observed in recent remote sensing research, where the conversion of built-up areas was associated with a decay in vegetation and fluctuations in the ecological conditions of cities.

Moreover, the better performance of EBBI in identifying built-up and barren surfaces was found to be consistent than other built-up index which showed that EBBI outperforms traditional built-up indices like NDBI and UI in diverse urban environments [4]. Previous GIS-based research also demonstrated the effectiveness of using multi-temporal NDVI for vegetation dynamic monitoring [14, 32] and recent studies highlighted the need for combining multiple spectral indices to achieve a more complete understanding of urban environmental dynamics. The present study is different from the previous studies which mainly focused on individual indices like NDWI, NDVI, or NDBI [1, 14, 17] as it uses eight complementary spectral indices for the simultaneous assessment of vegetation, water bodies, built-up land, and barren surfaces under both pre- and post-monsoon conditions. The multi-index framework is an integrated, comprehensive assessment of the seasonal land surface dynamics in Ara, a fast-expanding medium-sized city, for which similar geospatial studies are limited.

Positive NDVI values are typically linked with improved vegetation health, which can lead to lower urban heat island effect, better soil moisture storage, and ecological habitats [8, 33, 43]. In contrast, reduction in vegetation and addition of impervious surfaces are strongly associated with urban environmental degradation [35]. The same, it has been reported in previous studies that urbanization leads to a reduction in the area of rivers, wetlands and *khals* and rise in the risk of floods and water logging [24, 38]. The current work found similar spatial relationships among vegetation loss, urbanization growth, and changes in water-related variables. Although direct hydrological and

ecological observations were not the focus of this research, these relationships may be interpreted as statistical associations, not direct cause-effect relationships.

The results of this study have a relevance to sustainable urban planning and the management of wetlands in rapid urbanization cities like Ara. The spatially explicit framework of integrated use of NDVI, NDWI, MNDWI, NDBI, UI, NDBal, BI and EBBI facilitates the identification of vegetation loss, wetland degradation and built-up expansion areas. These can be used to help urban planners and local authorities consider the conservation of wetlands, control land reclamation and landfill operations, preserve urban green spaces, and integrate blue-green infrastructure into future development plans. In addition, frequent monitoring with multi-temporal satellite data can support policy makers to assess the success of restoration measures and develop evidence-based approaches for sustainable use of the land in growing cities like Ara. Recent research also has shown the potential of satellite-derived vegetation and urban indicators in the background of sustainable urban planning and climate adaptation [37].

Finally, the changes in the wetland-related indices observed could have wider ecological implications. Research has previously been conducted on the loss of migratory birds due to wetlands, mangroves and tidal flats degradation in South and Southeast Asia [40]. Field observations of bird abundance and biodiversity were not conducted in the present study, so it is not possible to make any direct inferences about migratory birds in Ara. Rather, these results highlight areas where further ecological research combining remote-sensing techniques and biodiversity sampling can benefit the understanding of how urban development affects wetland-breeding species. Additional ecological studies combining field studies and remote sensing are required to assess these potential effects (Figs. 5, 6, 7, 8 and 9).

7 Limitations

The objective of the study is to perform a detailed wetland, vegetation and urban surface dynamics analysis in Ara city, employing multi-temporal satellite images from the Landsat series and spectral indices. However, several limitations should be acknowledged.

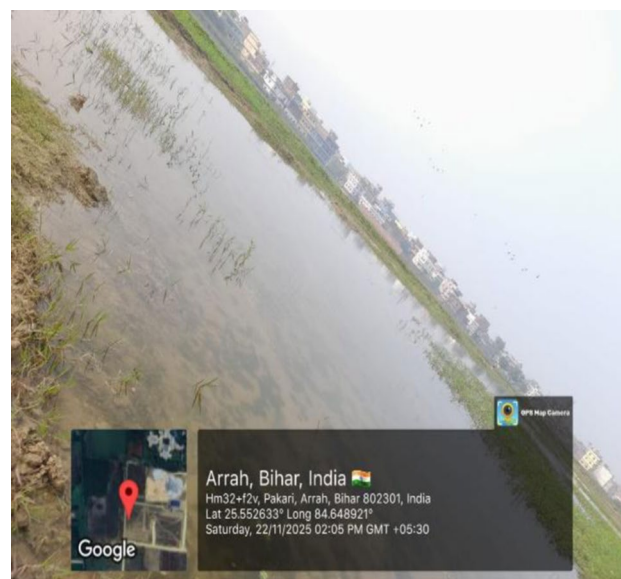


Fig. 5 Plate 1: Landfill and construction in Ara city (GPS Map Camera)



Fig. 6 Plate 2: Landfill and construction in Ara city (GPS Map Camera)

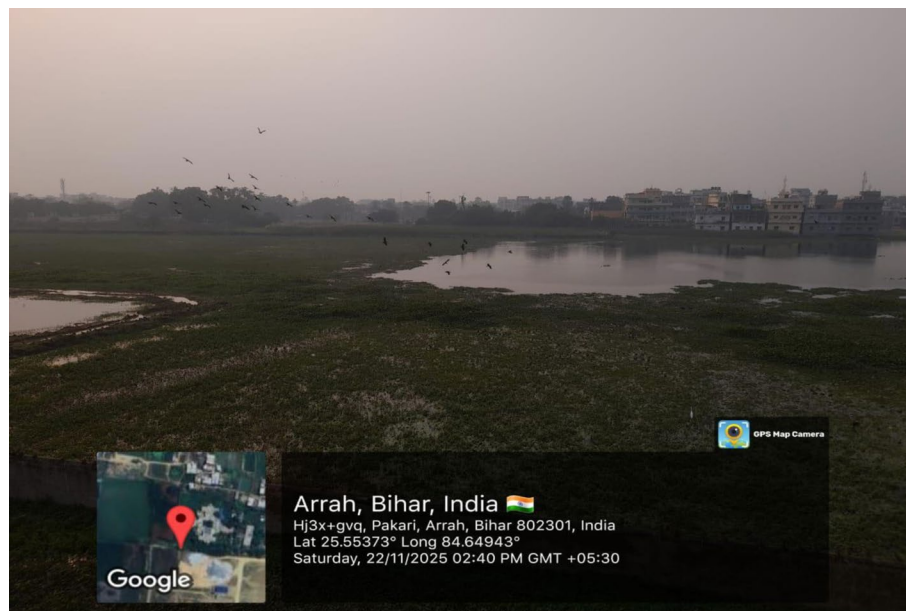


Fig. 7 Plate 3: Wetland plays key role for habitat and breeding of imigrant birds population (GPS Map Camera)

First, the analysis depend on completely on remotely sensed data, and there is no field verification of the wetland conditions, vegetation features or biodiversity. Second, the spatial resolution of Landsat data (30 m) could not always be resolved to fit the small size of wetlands or narrow vegetation patches, creating mixed pixel problems in heterogeneous urban features. Third, the inference of wetland dynamics is made from a spectral index instead of from direct measurements of water depth or water quality. Forth, the Landsat Collection 2 Level-2 images that were chosen was limited by data availability.

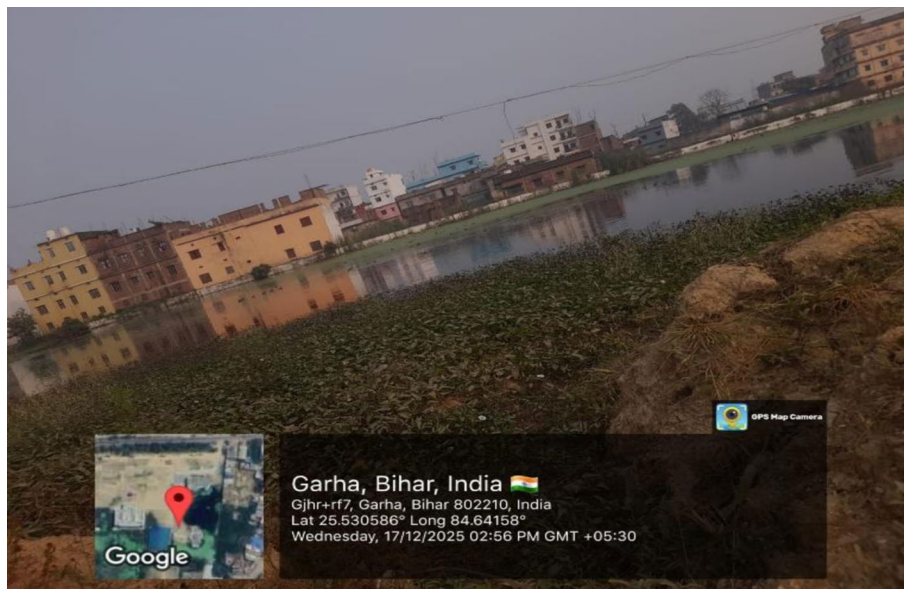


Fig. 8 Plate 4: Garha wetland, situated behind the Veer Kunwar Singh University (New Campus), Ara, showing urban development in close proximity to the wetland. The image illustrates potential anthropogenic pressure on urban wetlands, which may have ecological implications for habitat availability (GPS Map Camera)



Fig. 9 Plate 5: satellite image from Google Earth Pro

Satellite images were not available for the same calendar month throughout all study years, when cloud cover was over 10%. Hence, the analysis was carried out on the images taken during pre-monsoon (April–May) and post-monsoon (October–November) rather than any single month, to ensure the images were as comparable as possible and to minimize cloud contamination. In addition, while the ecological consequences of wetland alterations to migratory bird habitats are addressed, no field-based biodiversity surveys were carried out to confirm these conclusions. Future research should include high resolution satellite imagery, hydrological and water quality data, field observations and biodiversity surveys, in order to better understand long-term variations in wetlands and their influences on biodiversity. The integration of advanced machine learning algorithms and

socio-economic factors would further enhance urban environmental assessments and inform sustainable planning and conservation efforts.

8 Conclusions

This study evaluated the spatio-temporal variations of vegetation, wetlands, and built-up areas at Ara city from 1995 to 2025 using eight spectral indices extracted from multi-temporal Landsat images (NDVI, NDWI, MNDWI, NDBI, UI, NDBal, BI and EBBI). The outcomes show a clearly marked seasonal pattern of environmental conditions. The pre-monsoon NDVI values increased slightly throughout the study period while the post monsoon values decreased, suggesting poorer peak NDVI values and higher spatial variation in 2025 than in 1995. Pearson's correlation analysis also discovered significant negative associations between NDVI and all built-up indices ($p \leq 0.05$), indicating that built-up areas were associated with spatial variability of vegetation. The analysis of water-related indices indicates temporal changes in wetland and surface-water characteristics. The inverse relationship between the built-up indices and MNDWI as well as the contrasting behaviour of NDWI and MNDWI indicate changes in wetlands during the study period.

The combination of several spectral indices during the pre-monsoon and post-monsoon period serves as a holistic approach to monitoring long-term environmental change in a rapidly increasing urbanizing region. The outcomes could help urban planners and environmental managers determine where vegetation loss, wetland alteration, and urban development may be occurring, which could help guide sustainable land-use planning, wetland conservation, and ecosystem management. It is recognized that the changes observed in the wetlands could have ecological consequences for wetland dependent biodiversity, such as migratory birds, but the current study did not incorporate field-based biodiversity or data on bird populations. For this reason, these implications are preliminary and should be considered as yet to be validated with subsequent ecological studies. Integrating these techniques of remote sensing, land-use analysis, and field-based ecological surveys will give a more all-inclusive understanding of the relationships amongst urbanization, wetland dynamics, vegetation change, and the potential for preserving biodiversity in these wetlands.

Abbreviations

BI	Bareness Index
C.V.	Coefficient of variation
CO ₂	Carbon dioxide
EBBI	Enhance Built-up and Bareness Index
ESP	Electrostatic precipitator
GIS	Geographic Information System
GREEN _r	Green band reflectance
IBI	Index-based Built-up Index
L2C2	Level 2 Collection 2
LC	Landsat OLI/TIRS combined
LST	Land surface temperature
LT	Landsat Thematic Mapper
LULC	Land use land cover
MIR _r	Mid-infrared band reflectance
MNDWI	Modified Normalised Difference Water Index
MoEFCC	Ministry of Environment, Forest and Climate Change
MSS	Multispectral scanner system
NCERT	National Council of Educational Research and Training
NDBal	Normalized Difference Bareness Index
NDBI	Normalised Difference Built-Up Index
NDVI	Normalised Difference Vegetation Index
NDWI	Normalised Difference Water Index

NIR _r	Near Infrared band reflectance
NNE	North North-east
OLI	Operational Land Imager
RED _r	Red band reflectance
SDG	Sustainable development goal
St. Dev.	Standard deviation
SWIR1 _r	Short wave infrared band channel 1 reflectance
SWIR2 _r	Short wave infrared band channel 2 reflectance
TIRS1 _r	Thermal infrared sensor band channel 1 reflectance
TM	Thematic Mapper
UHI	Urban Heat Island
UI	Urban Index
USGS	United States Geological Survey
WNW	West North-west

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Author contributions

Both authors contributed to the study. The concept, draft preparation, the original manuscript writing, and the analysis in the GIS environment were carried out by Latifur Rahaman. Sadaf provided supervision, validation, and critical review of the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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